

Generation Interconnection System Impact Study Report

For Queue Project Q22-2029

Lafayette Road Station 138 kV

November 29, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2029. The Interconnection Customer proposes a 176.1 MW natural gas combined cycle facility interconnecting at the Lafayette Road Station 138 kV substation in the New Jersey area. The request was received on March 11, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 540 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 8 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$4,741,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The WISOM-SECORD 138.0 kV is overloaded to 127.5% of its 186 MVA emergency rating for the loss of the LA CYGNE-SECORD 138.0 kV. The project contributes 40.4% DFAX (71.2 MW) to this overload.

LAFAYETTE ROAD STA-SECORD 138.0 kV: 123.4% loading (365 MVA) following the outage of the CRAWFORDSVILLE POWER PLANT 345/138 kV XFMR; DFAX = 47.1%.

SECORD-LA CYGNE SUBSTATION 138.0 kV Line: 120.3% loading (210 MVA) following the outage of the WISDOM-SECORD 138.0 kV; DFAX = 40.9%.

For the loss of CAMP EVERS-LINDEN COGEN PLANT 138.0 kV, the PIPE-LINDEN COGEN PLANT 138.0 kV loads to 109.8% of its 393 MVA rating. Project DFAX: 17.4%.

PROFFIT'S SPRING-LA CYGNE 138.0 kV: 108.0% loading (195 MVA) following the outage of the WISDOM SUBSTATION-SECORD 138.0 kV; DFAX = 40.9%.

The LINDEN COGEN PLANT-CAMP EVERS 138.0 kV is overloaded to 107.7% of its 430 MVA emergency rating for the loss of the LINDEN COGEN PLANT-PIPE 138.0 kV line. The project contributes 17.8% DFAX (31.3 MW) to this overload.

WISDOM-THORNDIKE 138.0 kV Line: 106.7% loading (206 MVA) following the outage of the LA CYGNE-SECORD 138.0 kV line; DFAX = 40.4%.

ALDENE SUBSTATION-PINE TREE ACRES 138.0 kV: 105.9% loading (261 MVA) following the outage of the CAMP EVERS-LINDEN COGEN PLANT 138.0 kV; DFAX = 5.5%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
WISOM-SECORD 138.0 kV	Rebuild line with higher-capacity conductor	\$2,515,000	59% (\$1,495,000)
LAFAYETTE RD STATION-SECORD 138.0 kV	Rebuild line with higher-capacity conductor	\$2,446,000	50% (\$1,218,000)
LA CYGNE-SECORD SUBSTATION 138.0 kV	Rebuild line with higher-capacity conductor	\$1,771,000	59% (\$1,053,000)
PIPE-LINDEN COGEN PLANT 138.0 kV	Reconductor line	\$506,000	28% (\$144,000)
PROFFIT'S SPRING-LA CYGNE 138.0 kV line	Reconductor line	\$371,000	59% (\$221,000)
LINDEN COGEN PLANT SUBSTATION-CAMP EVERS SUBSTATION 138.0 kV	Reconductor line	\$456,000	27% (\$124,000)
WISOM-THORNDIKE 138.0 kV	Reconductor line	\$578,000	59% (\$343,000)
ALDENE SUBSTATION-PINE TREE ACRES 138.0 kV	Reconductor line	\$555,000	26% (\$143,000)
	Total allocated to project		\$4,741,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.